

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Final Report

Context to Consequence - AI-Driven News Categorisation and Event Prediction

by
Shuhan Liu

repository: <https://github.com/ese-ada-lovelace-2025/irp-sl1425.git>

Supervisors:
Mr Antony Sommerfeld; Dr Adriana Paluszny Rodriguez

August 2026

AI Acknowledgement Statement

I used Anthropic's Claude Sonnet 5 and Claude Haiku 4.5 via the Anthropic API (<https://www.anthropic.com/>) for selected code-development tasks and as part of the research pipeline. I used Claude Sonnet 5 for issue classification and used Claude Haiku 4.5 for stance and framing classification.

During the code-development process, I asked OpenAI's ChatGPT (GPT-5.6 Sol) (<https://chatgpt.com/>) and Claude Haiku 5 (<https://www.anthropic.com/>) for give me some advice on structuring of code pipeline. Both tools also was used to help me to organise implementation approaches, and explain some complex technical concepts for me. Additionally, I used both Claude and Codex to assist me with reproducibility, testing, debugging, and leakage checks. In addition, because there are so many code files that are combined in the early stages and subsequently method used in the final report, I asked ChatGPT and Claude to provide advice on the structure of the repository README, including the main workflow, run order, and reproducibility instructions. And I also ask them to give me some advice on figure and appendix structure.

For the final technical review of the repository, I used OpenAI Codex (<https://openai.com/codex/>) to help me to check the repository structure. It helped me to identify and clean up code files and notebooks from earlier stages of development that were no longer used in the final analysis. And it also helped me to refine the existing implementation and make the reproducibility process more robust.

I have checked all AI-assisted code and repository changes before they were in the final submission. I take full responsibility for the submitted work is my own. There are no generative AI are used to drafting, rewriting, and correction grammar to the submitted Project Plan or Final Report. And there are no generative AI used to generate figures and references.

Abstract

Rapid changes in party support make local election prediction difficult, especially for new parties. This study uses the 2026 Surrey local elections tests whether pre-election news adds predictive value beyond an adjusted history-only baseline, with particular attention to Reform UK. Stage 1 predicts candidate vote share from historical election results. Stage 2 models election-party residuals using party article share and stance balance. Stage 2 has v1 and v2 version, v2 added by-election data to v1. News-layer settings and predictions were fixed before final evaluation. Blinding applied only to the news-layer comparison. The final results showed no MAE improvement in v1, and v2 improved prediction only in some time windows. The clearest result was 31-90 time windows before the election. National and combined news reduced MAE by 0.2682 and 0.2404 percentage points, respectively. In exploratory analysis, combined news reduced MAE by 0.4948 points for non-Reform candidates but increased it by 0.6877 for Reform UK. Further analysis suggests that party differences and article volume explain much of the gain. Within-party stance had a smaller effect, but its direction was more consistent across windows. Overall, news effects varied by window, training data and party. Transfer results were mixed.

1 Introduction

1.1 Motivation and Research Question

UK election forecasts have used historical election results, party support and opinion polls (Stegmaier et al., 2023). However, subnational elections are harder to forecast because polls are often limited, electoral patterns differ across areas, and the competing parties can change across elections (Stoetzer et al., 2025). New parties present an additional challenge because they may have limited historical election data (Montalvo et al., 2019).

These limitations suggest a need for more timely information. But previous research provides mixed evidence on whether campaign media improves election forecasting (Bélanger & Soroka, 2012; UMEDA, 2023). Whether pre-election news adds predictive value beyond historical election information remains unclear. Hence, this study tests this question against an adjusted history-only baseline in Surrey local elections.

1.2 Approach and Objectives

This study uses a two-stage framework. Stage 1 builds a baseline from historical election data. Stage 2 uses structured news features to model the remaining prediction errors.

The study has three objectives: (1) establish a history-only baseline; (2) test whether news adds predictive value beyond a adjusted baseline; and (3) examine when news is useful and which factors affect its predictive value, with particular attention to Reform UK.

2 Literature Review and Research Positioning

2.1 Election Forecasting and Local Elections

At the subnational level, MRP combines polling, demographic and past-election information to estimate vote shares (Lauderdale et al., 2020). Another approach uses local-election results to forecast general elections, but failed to predict both vote shares and seat distributions in 2015 (Rallings et al., 2016). These studies show that information from different electoral

levels can support forecasting. However, information from one level may not always transfer well to another.

Another problem is limited polling at the subnational level. Stoetzer et al. (2025) use information from comparable subnational elections to support forecasting. In addition, Montalvo et al. (2019) combine pre-electoral survey data with national polls when new parties have little historical data. But each strategy relies on information that may not be available in every election.

2.2 News Coverage and Structured Text

Election-news studies provide mixed evidence on predictive value. Vermeer et al. (2024) found that media attention was associated with vote choice, especially for new parties. However, the effect of sentiment differed across media. Bélanger and Soroka (2012) found that campaign media tone did not explain errors in historical election forecasts. By contrast, UMEDA (2023) used media reports on district-level races to successfully predict the 2017 Japanese election, but these reports were based on extensive polling. These results show that media can be informative, but its predictive value beyond existing election information remains uncertain.

To use news in prediction, relevant articles must first be selected and converted into structured features. Wang et al. (2024) used LLM agents to select relevant news for time-series forecasting. In political news, Kuila and Sarkar (2024) showed that LLMs can convert unstructured political news into sentiment labels for specific political entities. Neither study shows that LLM-extracted news features can improve election forecasts.

2.3 Research Gap and Study Positioning

News may provide useful information, but some of this information may already be captured by existing election models. To address these problems, this study uses LLMs to filter and structure pre-election news and tests whether these features improve predictions beyond an adjusted no-news baseline.

3 Data and Feature Construction

3.1 Election Data

The core sample covers Surrey county-level elections in 2013, 2017, 2021 and 2026. The rolling design starts with 2013 as the initial training set, then uses 2017 and 2021 for sequential validation. The 2026 East and West Surrey elections form the final holdout. Borough and district records cover 11 councils and 46 election-year observations from 2018 to 2024. These records support local news searches and name standardisation but do not enter model training.

The study also includes 19 Surrey County Council by-elections from 2015 to 2026. Seventeen enter Stage 1, including eight that also enter Stage 2, eight used only in Stage 1, and Woking South as a separate blind case study. Of the remaining two, Warlingham remained sealed and unused, and Haslemere was later used for the replication test in Section 5.6.

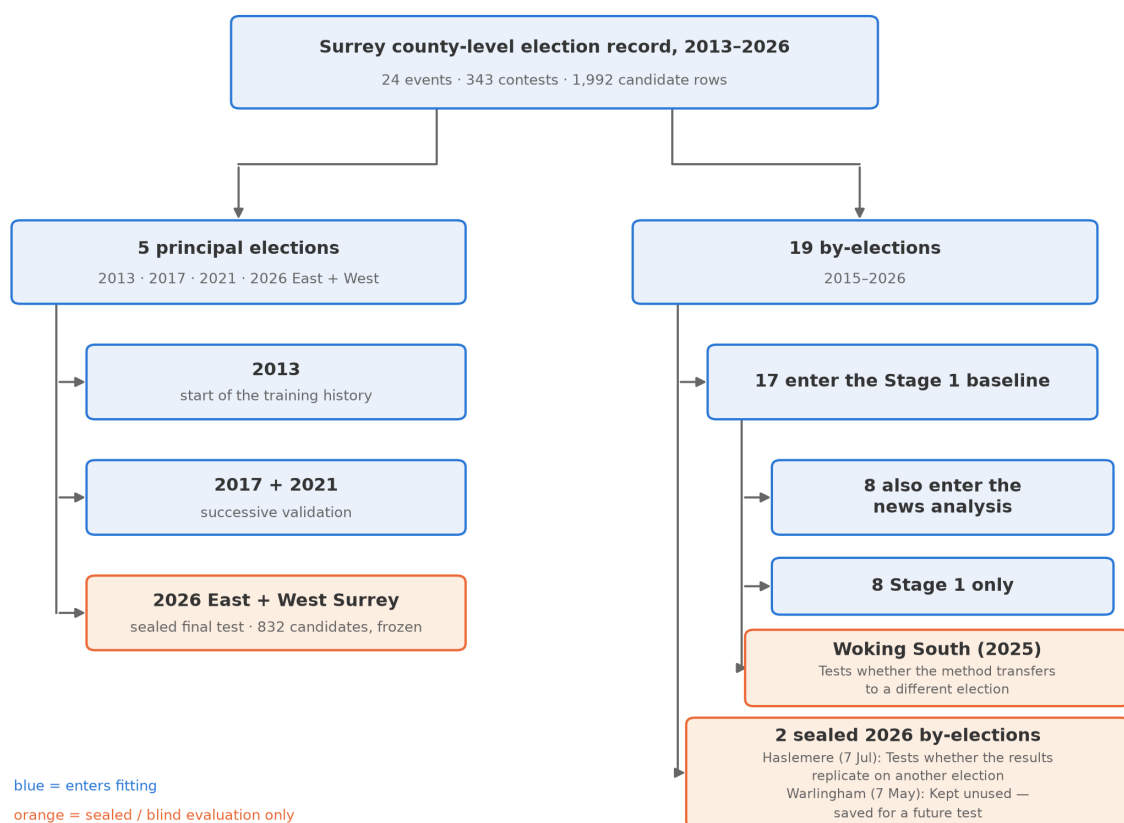


Figure 1: Role assignment of 24 county-level election events across the two-stage modelling framework and holdout evaluation (blue = included in fitting, orange = evaluation only).

3.1.1 Sources and unit of observation

The result of 2013, 2017 and 2021 main elections, 19 by-elections from 2015 to 2026, and separate 2026 East and West Surrey results are taken from official Surrey Council sources. Wikipedia provides the borough and district council results.

Across the 24 election events, the county-level dataset contains 343 electoral-area–election observations and 1,992 candidate records. These include 262 single-seat divisions and 81 two-seat wards. Each row represents one candidate in one electoral area and election. Candidate vote share equals the candidate’s votes divided by the total votes received by all candidates in that area.

3.1.2 Data standardisation

To keep records consistent across years, the study standardised candidate and party names for name order, character formatting and whitespace. The pipeline retained both original and standardised names and never merged records based only on name similarity. Reform UK and UKIP also remained separate throughout the pipeline. When a party did not contest an election, the dataset recorded its participation status rather than creating a zero-vote candidate row. If a party fielded multiple candidates in one area, each remained a separate record.

3.1.3 Validation and Source Tracking

The pipeline automatically validates the election dataset on every generated. It verifies candidate vote totals, compares recalculated vote shares with published results, and checks turnout, record consistency, seat counts and election dates. To allow for rounding on official pages, it accepts differences of up to 0.5 percentage points for vote share and turnout. The final quality report found no duplicate candidate IDs, identity mismatches, invalid vote shares or abnormal within-area totals. The pipeline also retains source and verification information. 1,971 of the 1,992 candidate records are provided by direct source links, and news archive pages were used to verify the remaining 21 records from four by-elections.

3.2 News Data and Corpus Construction

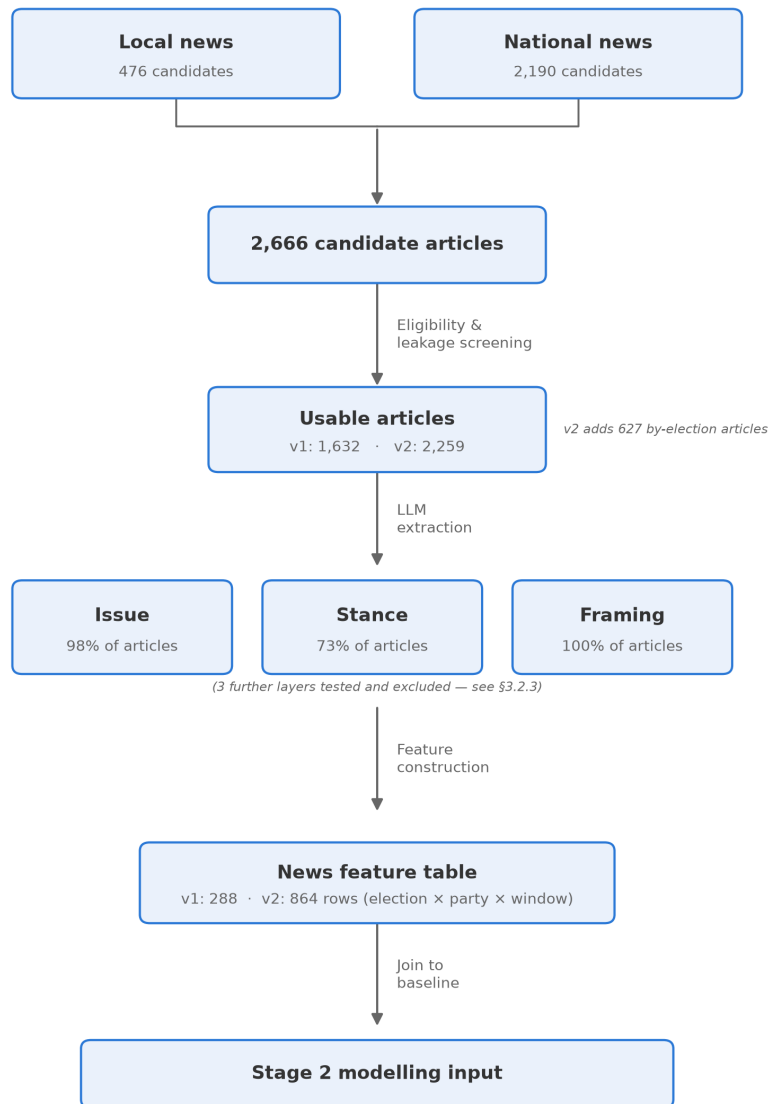


Figure 2: News data construction process. The diagram shows the v1 pipeline and the additional by-election articles included in v2, with further details in Section 3.2.1.

3.2.1 News sources and collection scope

The study treats Surrey local news and UK national political news separately. Local sources include SurreyLive, BBC News Surrey and Surrey-wide Guardian coverage, collected through search-engine APIs, publisher searches and web archives. National news comes from the Guardian Open Platform. Local searches cover 17 Surrey electoral areas selected under predefined rules, as shown in Table 1. The pipeline retained only verified geographic matches when electoral definitions changed across years.

To capture changes over time, the study uses six non-overlapping pre-election windows: 1–3, 4–7, 8–14, 15–30, 31–90 and 91–180 days. It also uses cumulative windows covering the previous 72 hours and 7, 14, 30, 90 and 180 days. Across the four principal elections, the corpus contains 1,632 articles: 188 local and 1,444 national. Adding 627 national articles from by-election periods brings the total to 2,259.

3.2.2 Eligibility and leakage screening

Before feature construction, the pipeline screened articles using predefined eligibility rules (Appendix Table 2). Each article required a reliable publication date, usable English text, genuine editorial content and publication within 180 days before the election. Polling-day articles were excluded because exact publication times were unavailable.

The pipeline also filter political relevance and result leakage. Local articles had to relate to the target area or Surrey politics, and national articles had to cover relevant UK politics. Any article containing results from the same election was excluded.

3.2.3 Structured news extraction and validation

The study used the Claude API to classify issue, stance and framing. During validation, it compared Claude Sonnet 5 and Claude Haiku 4.5 against a Cohen's kappa threshold of 0.60. The original stance and framing schemes fell below this threshold, so the study reduced them to three and four categories. Revised stance reached model agreement of 0.848 across 92 articles. Against human annotations, kappa was 0.741 for Haiku and 0.736 for Sonnet across 71 cases. The final pipeline used Sonnet 5 for issue and Haiku 4.5 for stance and framing. For framing, incumbent judgement and local impact met the threshold with limited evidence. Challenger emergence and voter discontent were too rare for reliable validation. The study excluded other failed tasks. Appendix Table 3 reports the detailed results.

The pipeline validated output format and evidence against source articles, allowing one retry for failures. It also reviewed 5% of eligibility decisions, including at least 30 articles per election. This left 1,604 of 1,632 articles with valid issue classifications, all 1,632 with framing labels, and 1,187 with stance labels because the remaining 445 mentioned no study party.

3.2.4 News feature construction

The study converted article-level outputs into features for 12 pre-election windows at the **election-party-period level**. For each party, it calculated two **stance-based features**: party article share and stance balance. Party article share measured the party's stance-labelled articles relative to all articles in the same election and window. Stance balance measured favourable minus unfavourable articles relative to all articles about that party. Neutral articles stayed in the denominator, affecting strength but not direction. Both features were calculated separately for local and national news and used in the final model.

Unlike stance, **article count**, **issue and framing** were calculated at the **election–window level**, so the same values applied to all parties within each election and window. The study used seven issue categories and four framing categories. Two issue columns were excluded after a mapping error left them empty.

4 Methodology

4.1 Prediction framework and research design

Using news to predict outcomes directly would make it difficult to separate its incremental contribution from the electoral structure represented in historical data. Therefore, this study used a two-stage design, with Stage 1 modelling the historical baseline and Stage 2 testing whether news explained the remaining residuals.

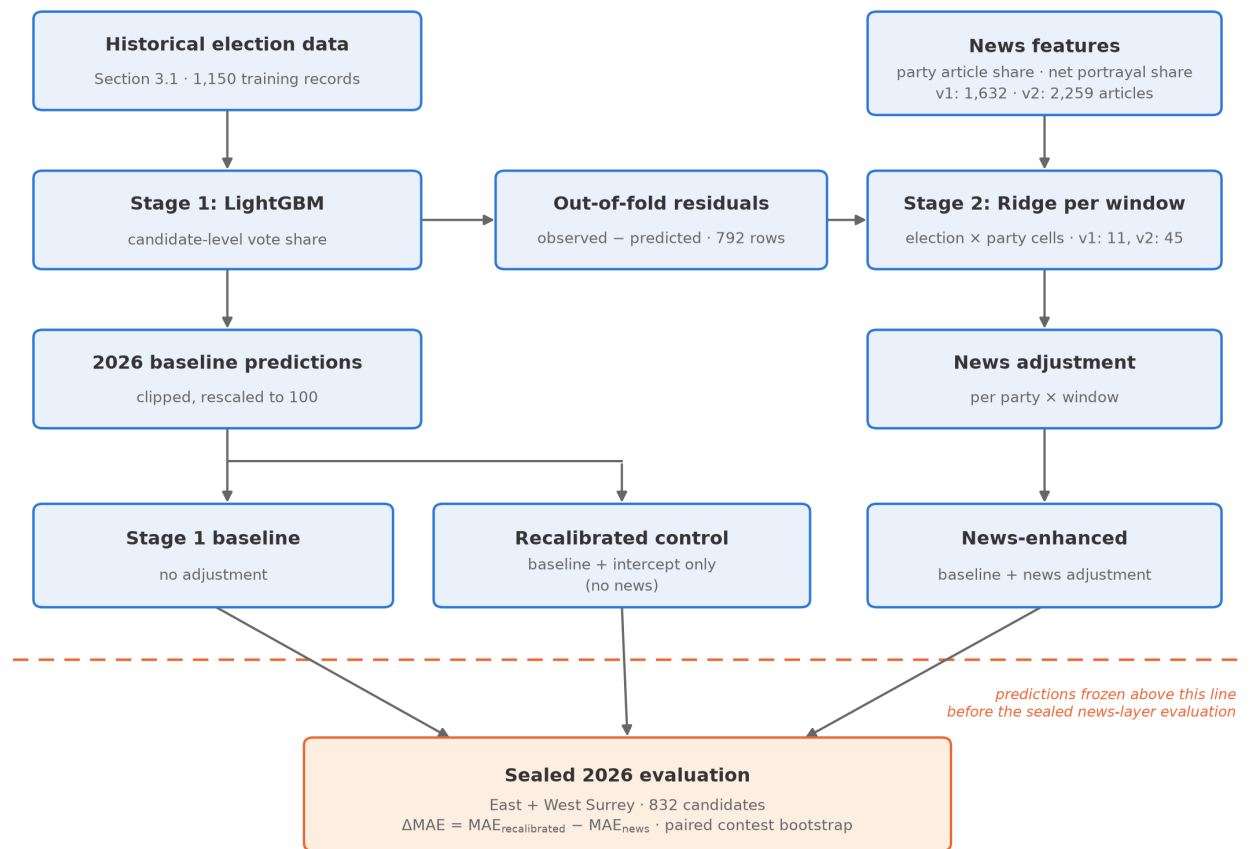


Figure 3: Two-stage prediction framework. Stage 2 adjusts the Stage 1 baseline using news, and compares it with a no-news adjustment control. Predictions above the dashed line were frozen before the final news-layer evaluation.

Stage 2 averaged residuals by election and party and fitted a separate model for each pre-election window. As an exploratory check, the study compared this residual design with a direct model combining historical and news features on pre-2026 validation data. The residual design performed better in 16 of 18 comparisons (Appendix Table 4).

4.2 Stage 1: Historical election baseline

Stage 1 predicts candidate vote share from historical elections, with one row per candidate and contest. All pre-2026 contests were single-seat divisions, while the 2026 holdout used two-seat wards. As a result, average candidate share fell from 22.65% to 9.75%. To account for differences in candidate count across contests, the target was candidate vote share $\times$ candidate count / 100. This gives each contest a mean target of 1, and predictions were then converted back to vote share.

Stage 1 used no news features. Each candidate record was represented by 35 predictors describing electoral history and contest context. After preprocessing and encoding, these became 127 model inputs (Appendix Table 5). Within each fold, all preprocessing statistics were learned only from the training data to prevent leakage.

For model selection, the study compared Ridge regression, partial pooling and LightGBM (Ke et al., 2017) to represent regularised linear, pooled and flexible tree-based approaches. Four temporal folds covered 2017, 2021 and two 2025 by-election periods, with training restricted to earlier elections. Because Reform UK had little historical data, the predefined model-selection rule focused on Reform UK performance. Across 14 Reform UK validation cases, LightGBM reduced MAE by 8.3% relative to Ridge, exceeding the 5% threshold, and was selected. Across all candidates, out-of-fold MAE was 9.85 for LightGBM, 8.97 for Ridge and 8.87 for partial pooling.

After model selection, Stage 1 produced out-of-fold predictions for 792 candidate records across 16 time-based folds (two general-election polling days and 14 by-election polling days). Each fold corresponded to one polling day. The resulting residuals (observed vote share minus predicted vote share) formed the Stage 2 training targets. The final LightGBM model used 1,150 candidate records from elections before 7 May 2026 to predict the 2026 holdout. Within each contest, negative predictions were set to zero and the rest rescaled to sum to 100.

4.3 Stage 2: Modelling incremental news information

Stage 2 modelled Stage 1 residuals, with positive values indicating underestimation. News features varied by election and party rather than by candidate, so residuals were averaged within each election and party cell. The analysis covered the Conservatives, Labour, Liberal Democrats, Green Party, UKIP and Reform UK.

Based on this structure, the Stage 2 training sample expanded under v2. Under v1, it contained 11 election-party cells from 2017 and 2021. 2013 served as the initial Stage 1 training election and therefore had no out-of-fold residuals. Adding eight by-elections under v2 increased the total to 45 cells and the number of Reform UK cells to seven, compared with only one in 2021. Cells with no articles were retained, with their news features set to zero.

For each news type, Stage 2 used party article share and stance balance (Section 3.2.4). The main analysis used combined and national news separately. Local news was used in the sensitivity analysis. Each of the 12 windows had a separate Ridge model. Because of small training set, so ridge provided simple regularisation against overfitting, with $\alpha = 1.0$ fixed before validation. The study standardised predictors using the training mean and standard deviation, and subtracted the mean training residual from the target before fitting.

$$\hat{r}_{ep} = \bar{r} + \sum_{j=1}^2 \beta_j \frac{x_{epj} - \bar{x}_j}{s_j}.$$

Here, $\hat{r}_{ep}$ is the predicted residual for election e and party p , and $\bar{r}$ is the mean training residual. The Ridge model learned how each news feature was associated with Stage 1 residuals in the training cells. Each coefficient β_j measures the change in the residual associated with a one-standard-deviation increase in the corresponding news feature.

Later, stage 2 added each predicted residual to the corresponding Stage 1 prediction. All candidates in the same election–party–period group received the same adjustment. Negative predictions were then set to zero and each contest rescaled to sum to 100.

4.4 Leakage Control and Blinding

The modelling process followed time order, and model features used only information available before each target election. To reduce leakage risk, the pipeline also kept model features and election outcomes in separate files.

Although the 2026 articles were collected after the election, only articles published before polling day were used to construct the news features.

Before the final news-layer evaluation, the v1 and v2 predictions were frozen, with the v2 training composition fixed in the same protocol (Appendix Table 16). During Stage 1 development, performance metrics on the 2026 holdout had already been viewed, but the 2026 outcomes were not used for training or model selection. The local-news search areas (Table 1) used pre-registered rules and publicly available 2026 outcomes before examining any news content. Hence, blinding applied only to the final news-layer comparison. The full pipeline was not outcome-blind.

4.5 Evaluation Strategy

The main evaluation covered 832 candidate predictions across the East Surrey and West Surrey elections. The vote-share analysis used 753 candidates from the Stage 2 party groups, excluding 79 candidates from other parties or independents. Candidate vote-share MAE was the primary metric, with RMSE used to capture larger errors. Seat-call accuracy was a secondary outcome, with the top two predicted candidates in each ward treated as elected.

To isolate the value of news from simple adjustment, the study compared each news-enhanced prediction with a no-news adjustment control. This control used the mean training residual as the Stage 2 intercept and added it to the Stage 1 baseline. It then applied the same clipping and within-contest normalisation. The study defined incremental news value as

$$\Delta\text{MAE} = \text{MAE}_{\text{adjustment}} - \text{MAE}_{\text{news}}.$$

A positive value meant that news reduced prediction error.

The frozen protocol defined combined and national news across the six non-overlapping windows as the primary confirmatory family, giving 24 comparisons. Local news, cumulative windows and the 2017-only variant were pre-specified as sensitivity analyses. Uncertainty was estimated using a pairs cluster bootstrap with 2,000 resamples, treating the 81 contests as the resampling units (MacKinnon et al., 2023). Stage 2 was fitted on 45 election–party cells (Section 4.3), and candidates within the same cell shared the same news adjustment. Therefore, candidate-level observations were not treated as independent, and uncertainty was estimated at the contest level. This approach was appropriate because the estimated quantity was the difference in MAE rather than a regression coefficient, and the number of clusters (81) was reasonably large (MacKinnon et al., 2023). Each resample compared the two models on

the same contests, and the resulting differences were used to construct a 95% confidence interval.

4.6 Diagnostic and Robustness Tests

In addition to the confirmatory analysis, the study evaluated three pre-specified sensitivity settings, including local news, cumulative windows and 2017-only training. LOPO was conducted after the final news-layer evaluation as a diagnostic robustness check. LOPO removed each of the five parties from the 2017 training data in turn, refitted the model on the other four, and tested performance on 2021 data for single-party dependence.

The study also checked whether news features had enough variation for reliable analysis. A feature needed at least three distinct training values for model fitting and at least ten for reported confirmatory analysis. The election-level issue and framing features had four distinct values. Therefore, they passed the fitting threshold but not the reporting threshold, so the study excluded them from the reported confirmatory models.

After the final news-layer evaluation, the study used exploratory diagnostics to examine other explanations for the main findings. Party-indicator models tested whether fixed party differences explained the improvement. Within-party stance analysis removed each party's average stance within the same training window. And this study also used volume-only control tested whether the signal reflected the amount of coverage rather than its content. Under the pre-declared rule, the study kept all sensitivity and exploratory results separate and did not promote them to confirmatory findings.

5 Results

5.1 Main Confirmatory Results

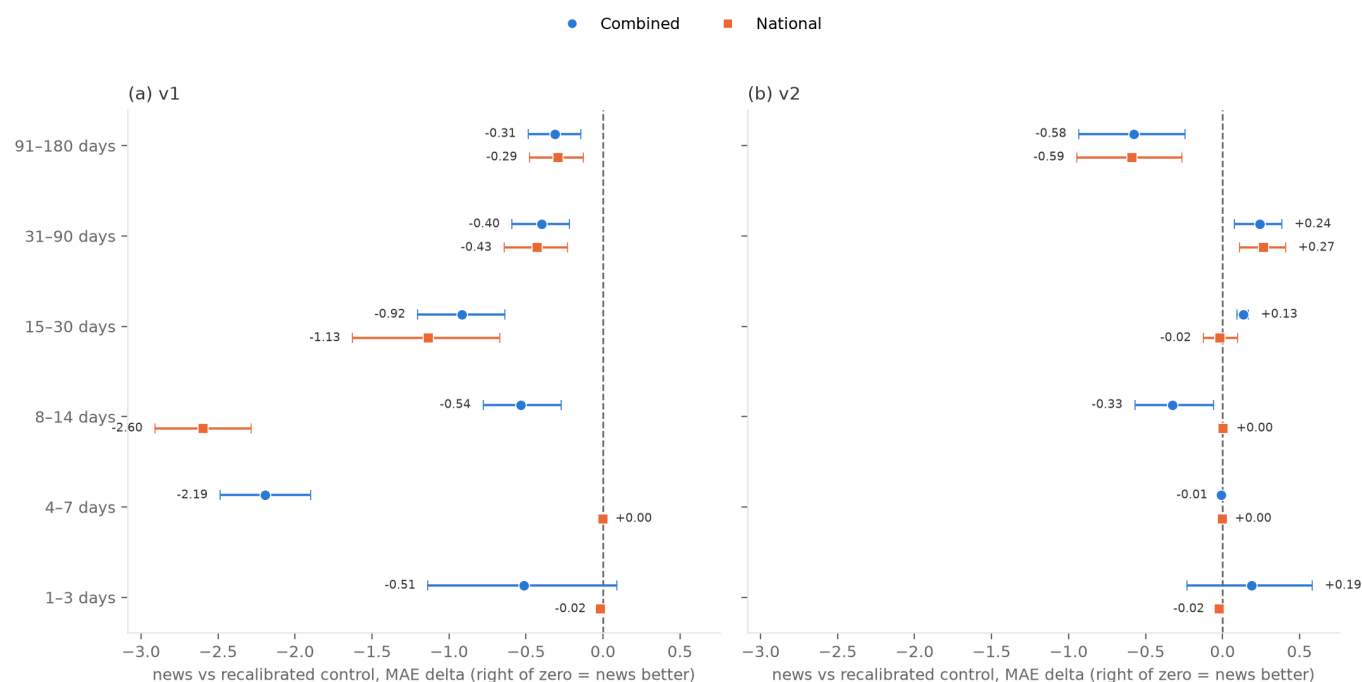
The confirmatory evaluation on the 2026 holdout covered 753 candidate predictions from the Stage 2 party groups. The Stage 1 historical baseline achieved an MAE of 4.4410 percentage points. After the no-news residual adjustment, MAE was 4.4414 under v1 and 4.4454 under v2.

The strongest confirmatory evidence came from v2. Five of its 12 specifications reduced MAE, and four had 95% bootstrap confidence intervals entirely above zero (Figure 4; Appendix Tables 6–7). In contrast, v1 produced no MAE improvements, and most 95% confidence intervals lay entirely below zero, this means worse performance than the no-news control. The difference between v1 and v2 reflects changes introduced in v2, and their individual effects are examined later.

The clearest improvement appeared in the 31–90 day window. Combined news reduced MAE from 4.4454 to 4.2051, an improvement of 0.2404 percentage points [0.0781, 0.3866]. National news reduced MAE to 4.1773, an improvement of 0.2682 [0.1088, 0.4110]. This was the only v2 window where both news types had positive confidence intervals. The main negative result appeared at 91–180 days, where both news types worsened MAE. Hence, news improved prediction in several v2 time-windows.

5.2 Sensitivity to Training Design and News Specification

Sensitivity was assessed by checked varied the construction of the news signal and check examined training composition.



Whiskers: contest-bootstrap 95% intervals (2,000 resamples). Source: unblinding record, register section 16.

Figure 4: Forest plot of the 24 confirmatory comparisons. Positive values indicate lower MAE with news. No v1 window showed positive confidence intervals for both news types, while 31–90 days was the only v2 window where both did.

Using local news only, v2 continued to reduce MAE at key windows. At 31–90 days, MAE fell from 4.4454 to 4.2571, with additional improvements at 8–14 and 91–180 days. The largest gain occurred at 91–180 days (+0.2894), showing that the timing of the strongest local-news signal differed from the main results. (Appendix Table 9).

The window coverage also affected the strength of the signal. Under v2, combined news reduced MAE across the shorter cumulative windows. However, the 180-day combined window moved in the opposite direction. The largest improvement occurred over the previous 14 days, where MAE fell from 4.4454 to 3.8353 and RMSE from 5.7538 to 4.8741. Local news also showed improvements over shorter cumulative windows (Appendix Table 10).

In contrast, the signal was more sensitive to training design. The 2017-only setting contained five training cells and improved MAE in 0 of 18 comparisons, reproducing the v1 pattern (Appendix Table 8). In the leave-one-party-out analysis, removing the Conservatives produced improvements in 6 of 18 comparisons, but removing any of the other four parties produced none (Appendix Table 11).

5.3 Alternative Explanations for the News Signal

In the v2 31–90 day window, party indicators improved MAE by 0.8342 percentage points, compared with 0.2404 for the frozen combined-news specification. Adding news to the party-indicator model increased the improvement to 0.8504. Party indicators also outperformed frozen combined news across all six non-overlapping windows. However, within-party stance had the same sign as the frozen combined-news effect in five of the six windows, with the sixth being zero. The constant +0.8342 party-indicator effect matched the sign in only three. These results suggest that fixed party differences explain much of the overall improvement, but

within-party stance more closely followed changes in the news effect across windows.

The study next examined whether the signal could be explained by reporting intensity. In the 31–90 day window, article volume alone improved MAE by 0.2247 percentage points. Combining volume and stance increased the improvement to 0.3413. Across the six non-overlapping windows, stance alone outperformed volume alone in five, with an average advantage of 0.0993 percentage points in MAE improvement. This suggests that stance contained predictive information beyond reporting intensity alone.

The study then tested whether stance remained informative after removing average party differences. Within-party stance improved MAE by 0.2291 percentage points in the 31–90 day window [0.1669, 0.2876]. After the final evaluation, seven issue and framing features fixed before fitting were also explored. Incumbent-judgement framing improved MAE by 0.5718 and was the only feature whose 95% confidence interval lay entirely above the 0.3413 volume-plus-stance improvement (Appendix Table 15). However, only two windows had sufficient training variation for these features, so the results remained exploratory. Overall, party structure and article volume explained important parts of the signal, but within-party stance and framing still showed predictive value when tested separately.

5.4 Reform UK and Model Behaviour

Reform UK had limited historical representation in v2, with 12 candidate-level observations compared with 65–168 for the other five parties. Even so, Stage 1 achieved a Reform MAE of 3.2318 in 2026, below the overall MAE of 4.4410.

After adding news, the effect differed sharply between Reform and non-Reform candidates. In the 31–90 day combined specification, ΔMAE was +0.4948 for non-Reform candidates but -0.6877 for Reform UK candidates, while the overall gain of +0.2404 was a candidate-count-weighted average that masked this difference (Figure 5). The direction of the adjustment was consistent across elections, but its effect depended on the starting baseline. In 2021, Reform's signed error relative to the recalibrated control fell from +12.21 to +3.07, correcting much of the overprediction. In 2026, however, it moved from -1.33 to -3.16 , increasing the existing underprediction.

The difference between v1 and v2 was partly associated with the inclusion of Reform UK training cells. The 2021 validation setting contained no Reform UK training cells, whereas v2 included seven Reform UK cells. Removing these cells reduced the combined-news gain from +0.2404 to +0.0773 and the national-news gain from +0.2682 to -0.0015 (Appendix Table 12). These results suggest that Reform UK training cells contributed to how the news adjustment generalised to the 2026 holdout.

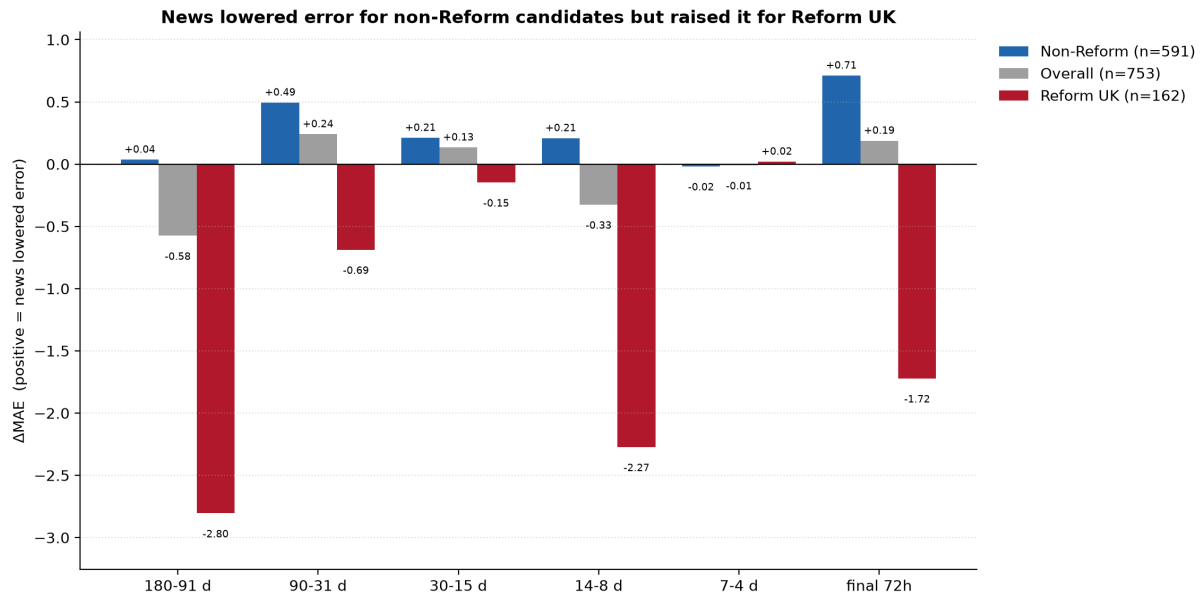
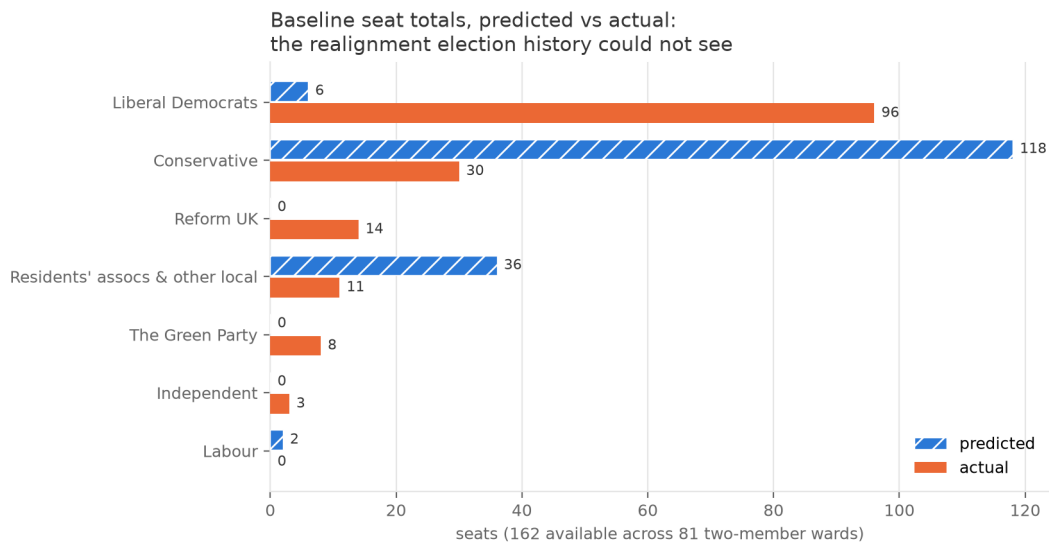


Figure 5: Post-evaluation exploratory decomposition of ΔMAE after news adjustment by party group and prediction window in the 2026 holdout under the combined-news specification. Bars show non-Reform, overall, and Reform UK candidates. Positive values indicate reduced error after the news adjustment, with overall values calculated as candidate-count-weighted averages.

5.5 Seat-level prediction results

Seat-call accuracy was a secondary outcome. Although Stage 1 achieved an accuracy of 0.7188, it failed to capture the 2026 seat-level changes. It correctly predicted both winning candidates in only 18 of 81 wards (Figure 6).



Source: Stage 1 holdout predictions vs observed results; register section 16 addendum.

Figure 6: Stage 1 predicted and observed seat totals in 2026. The largest differences were Conservatives 118 vs 30, Liberal Democrats 6 vs 96, and Reform UK 0 vs 14, showing that the historical baseline missed the 2026 seat realignment.

The simple ‘elect-nobody’ baseline classified every candidate as not elected and achieved 0.8053 accuracy. The four confirmed vote-share specifications remained below this benchmark for seat-call accuracy (Appendix Table 13). The form shows seat-call accuracy did not align with confirmed MAE improvement. This suggests that news affected vote-share error and seat-call accuracy through different specifications. As seat-call accuracy was secondary, these results provide supplementary evidence.

5.6 Transfer and replication tests

For Woking South, the pre-registered transfer rule was applied to the only informative window (91–180 days). Local news was selected within this window based on the highest 2026 Δ MAE. But this specification was the worst of the 18 candidate specifications on transfer, with a Δ MAE of -3.80 relative to the no-news baseline and increased error across all five parties (Appendix Table 14). The transfer rule was therefore not supported.

Haslemere showed the same window pattern as the main 2026 results, with 31–90 days performing best and 91–180 days performing worst. Three of the 12 frozen v2 specifications had lower MAE than the no-news adjustment control (Appendix Table 14). The performance of individual specifications varied across tests. However, the mid-range window again performed relatively well in a new unseen election.

6 Discussion

6.1 Interpreting the Incremental Predictive Value of News

The frozen confirmatory analysis showed that pre-election news added predictive value under specific conditions. All confirmatory improvements occurred in v2, with the strongest evidence in the 31–90 day window. In this window, both combined and national news outperformed the no-news adjustment control, with 95% bootstrap confidence intervals entirely above zero.

As shown in Section 5.4, the v2 gains were sensitive to training composition. Excluding the seven Reform UK training cells made the gains much smaller, especially for national news. The predictive value of news therefore depended on both timing and the composition of the training data.

This conditional pattern is consistent with the mixed evidence in previous election forecasting research (Bélanger & Soroka, 2012; UMEDA, 2023). These studies suggest that the predictive value of media information may depend on what the media measure captures and what information is already captured by the baseline model.

6.2 What the News Signal Represents

The results suggest that some information in the news features overlapped with party identity. Removing the average residual for four established parties increased the frozen-news improvement from 0.2404 to 0.3004 percentage points. However, removing the average residual for all six parties reduced it to 0.0428, with the confidence interval crossing zero $[-0.0661, 0.1461]$. This is consistent with the finding of Titelman and Lauderdale (2025) that candidate positions and party labels can act as substitutes in explaining vote choice. This provides a parallel to the overlap between news features and party identity found here.

Party identity did not explain everything. Article volume and within-party centred stance each improved prediction when tested separately. Vermeer et al. (2024) also examined me-

dia visibility and sentiment separately. The study found positive visibility effects, but sentiment effects differed across media. In this study, article volume and stance balance were strongly correlated in this study ($r = -0.778$). After party identity and article volume were included, adding centred stance did not improve prediction ($\Delta\text{MAE} = -0.1229$, 95% CI $[-0.1595, -0.0879]$). Thus, stance improved prediction when tested separately, but did not add predictive value beyond party identity and article volume.

These results also highlight that news effects may vary across parties. As shown in Section 5.4, the same adjustment that improved non-Reform predictions increased Reform error. This is consistent with Vermeer et al. (2024), who found stronger visibility effects for new parties than for established parties. Average party-level relationships may not generalise equally across parties.

6.3 Dependence on Training Design and Baseline Error

The v2 expansion added more elections and residual patterns to Stage 2. The added by-election cells were less reliable on their own, with estimated cell-level reliability falling from 0.9375 to 0.2253. However, the estimated correlation needed for detection changed very little, from 0.6218 to 0.6193. This suggests that the larger training sample largely made up for the lower reliability of the added cells.

Training composition also mattered. The LOPO results changed when different parties were excluded (Section 5.2), and removing the seven Reform UK training cells substantially reduced the news improvement (Section 5.4). Stoetzer et al. (2025) show that pooling information across comparable subnational elections can improve forecasting when election-specific data are sparse. The results here extend this point by showing that the benefit also depends on which party and election cases are included in training.

The news adjustment also depended on the baseline error. The same downward adjustment reduced Reform UK overprediction in 2021 but increased underprediction in 2026 (Section 5.4). Because stage 2 learns from past prediction errors. If the baseline error changes direction in a new election, the same correction can make the prediction worse.

6.4 Methodological Implications and Limitations

The news content covered both national and local reporting, but geographic detail remained limited. Only 68 of 1,632 articles could be linked to a single Surrey division, so division-level news features were not used. This is consistent with earlier text-based election forecasting research highlighting the importance of geographic resolution (Burnap et al., 2016). As noted in Section 4.4, the local search areas were selected using 2026 election outcomes. Local articles represented 8.3% of the v2 corpus. The local-only specification was used only in sensitivity analysis, although local articles also contributed to the combined specification. Therefore, results involving local news and the local component of combined specifications should be interpreted more cautiously.

Issue and framing features also had limited training variation. They passed the fitting threshold but not the reporting threshold (Section 4.6), so their positive exploratory results require further testing.

The confirmatory analysis involved 24 comparisons without formal multiplicity correction, meaning some positive findings could occur by chance (Liu & Shiraito, 2023). The comparison structure was pre-specified, but no single window was designated as primary. The 31–90 day window was the only v2 window positive for both combined and national news, but it remained

part of the 24 comparisons. Transfer results were mixed, so further testing across other areas and elections is needed.

7 Conclusion

This study established a history-only baseline and tested whether pre-election news added predictive value beyond a no-news adjustment control. Under v2, several specifications reduced MAE. The clearest improvement appeared in the 31–90 day window, where both combined and national news reduced MAE. v1 did not show the same improvement. The v2 results should still be interpreted with caution because 24 comparisons were tested and transfer performance was mixed.

Further analysis showed that several factors contributed to the predictive gain. Party identity and article volume explained much of the improvement. Centred stance also improved prediction when tested separately. Within-party stance showed a more consistent direction across windows. However, it did not add predictive value after party identity and article volume were included. The effect also differed across parties. In the 31–90 day combined-news analysis, prediction improved for non-Reform candidates but worse for Reform UK.

Overall, pre-election news added predictive value under some settings. Its value changed across time windows and parties, and it also depended on the training data. The study also developed an auditable two-stage forecasting framework. Future work should improve local news coverage and geographic detail.

References

- Bélanger, É., & Soroka, S. (2012). Campaigns and the prediction of election outcomes: Can historical and campaign-period prediction models be combined? *Electoral Studies*, 31(4), 702–714. <https://doi.org/10.1016/j.electstud.2012.07.003>
- Burnap, P., Gibson, R., Sloan, L., Southern, R., & Williams, M. (2016). 140 characters to victory?: Using twitter to predict the uk 2015 general election. *Electoral Studies*, 41, 230–233. <https://doi.org/10.1016/j.electstud.2015.11.017>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30. https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html
- Kuila, A., & Sarkar, S. (2024, May). Deciphering political entity sentiment in news with large language models: Zero-shot and few-shot strategies. In H. Afli, H. Bouamor, C. B. Casagran, & S. Ghannay (Eds.), *Proceedings of the second workshop on natural language processing for political sciences @ Irec-coling 2024* (pp. 1–11). ELRA; ICCL. <https://aclanthology.org/2024.politicalnlp-1.1/>
- Lauderdale, B. E., Bailey, D., Blumenau, J., & Rivers, D. (2020). Model-based pre-election polling for national and sub-national outcomes in the US and UK. *International Journal of Forecasting*, 36(2), 399–413. <https://doi.org/10.1016/j.ijforecast.2019.05.012>
- Liu, G., & Shiraito, Y. (2023). Multiple hypothesis testing in conjoint analysis. *Political Analysis*, 31(3), 380–395. <https://doi.org/10.1017/pan.2022.30>
- MacKinnon, J. G., Nielsen, M. Ø., & Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2), 272–299. <https://doi.org/10.1016/j.jeconom.2022.04.001>
- Montalvo, J. G., Papaspiliopoulos, O., & Stumpf-Fétizon, T. (2019). Bayesian forecasting of electoral outcomes with new parties' competition. *European Journal of Political Economy*, 59, 52–70. <https://doi.org/10.1016/j.ejpoleco.2019.01.006>
- Rallings, C., Thrasher, M., & Borisyuk, G. (2016). Forecasting the 2015 general election using aggregate local election data. *Electoral Studies*, 41, 279–282. <https://doi.org/10.1016/j.electstud.2015.11.007>
- Stegmaier, M., Jokinsky, S., & Lewis-Beck, M. S. (2023). The evolution of election forecasting models in the uk. *Electoral Studies*, 86, 102694. <https://doi.org/10.1016/j.electstud.2023.102694>
- Stoetzer, L. F., Erfort, C., Rajski, H., Gschwend, T., Munzert, S., & Koch, E. (2025). An election forecasting model for subnational elections. *Electoral Studies*, 95, 102939. <https://doi.org/10.1016/j.electstud.2025.102939>
- Titelman, N., & Lauderdale, B. E. (2025). The effects of party labels on vote choice with realistic candidate differentiation. *Political Science Research and Methods*, 13(3), 591–610. <https://doi.org/10.1017/psrm.2024.20>
- UMEDA, M. (2023). Aggregating qualitative district-level campaign assessments to forecast election results: Evidence from japan. *International Journal of Forecasting*, 39(2), 956–966. <https://doi.org/10.1016/j.ijforecast.2022.03.006>
- Vermeer, S., Van Remoortere, A., & Vliegthart, R. (2024). Still going strong? the role of traditional media in the 2021 dutch parliamentary elections. *Acta Politica*, 59(3), 519–535. https://pure.uva.nl/ws/files/181373203/Still_going_strong.pdf
- Wang, X., Feng, M., Qiu, J., Gu, J., & Zhao, J. (2024). From news to forecast: Integrating event analysis in LLM-based time series forecasting with reflection. *Advances in Neural Information Processing Systems*, 37, 58118–58153. <https://doi.org/10.52202/079017-1853>

A Local News Search Areas

electoral area	pre-registered selection reason
Addlestone Ward	strongest 2026 Reform UK share (17.0%)
Ashted Ward	winning party changed 2021–2026
Bagshot, Windlesham & Chobham Ward	winning party changed 2021–2026
Banstead, Woodmansterne & Chipstead Ward	safe seat (2021 margin 52.8pp)
Caterham Hill Ward	marginal seat (2021 margin 2.1pp); winning party changed 2021–2026
Epsom Town & Downs Ward	marginal seat (2021 margin 0.1pp); winning party changed 2021–2026
Ewell Court, Auriol & Cuddington Ward	safe seat (2021 margin 52.4pp); winning party changed 2021–2026
Farnham Central Ward	safe seat (2021 margin 44.8pp)
Guildford East	marginal seat (2021 margin 1.8pp)
Horley West, Salfords & Sidlow Ward	strongest 2026 Reform UK share (17.0%)
Lingfield Ward	strongest 2026 Reform UK share (19.0%)
Lower Sunbury & Halliford Ward	marginal seat (2021 margin 0.7pp)
Redhill East	safe seat (2021 margin 44.3pp)
Shere	marginal seat (2021 margin 1.8pp)
Stanwell, Stanwell Moor & Ashford North Ward	strongest 2026 Reform UK share (20.0%)
Tadworth, Walton and Kingswood	safe seat (2021 margin 58.9pp)
Thorpe, Longcross & Ottershaw Ward	strongest 2026 Reform UK share (17.0%)

Table 1: The 17 Surrey electoral areas included in the local news search sample and their pre-registered selection reasons. Areas were selected across four electoral-structure categories: safe seats (2021 winning margin ≥ 20 percentage points), marginal seats (≤ 5 percentage points), areas where the winning party changed between 2021 and 2026, and areas with the highest Reform UK vote shares in 2026. The top five areas in each category were selected and then deduplicated. The rules and thresholds were fixed before any news content was examined, and the selection used only publicly available election-result data. An area could satisfy more than one criterion.

B Article Eligibility Rules

code	inclusion criterion (all must hold)	applied by
I1	Confirmed or Probable publication date within the 180-day window ($1 \leq \text{day_index} \leq 180$), or polling day with a confirmed pre-07:00 timestamp	script
I2	Published before voting began: nothing at or after 07:00 on polling day	script
I3	Relevant to the arm collected: passes a Local (L1–L4) or National (N1–N3) test	human
I4	Genuine editorial content: report, analysis, opinion, editorial, letter, interview, profile, or published press release	human
I5	English language	script
I6	Minimally retrievable: headline, source, date, and an extract or full text lawfully obtainable	script
code	exclusion rule (checked in order; first failure logged)	applied by
E1	Date unresolved — no publication date recoverable at Probable or better confidence	script
E2	Outside window — more than 180 days before, or dated after, polling day	script
E3	Voting under way — published at/after 07:00 on polling day, or polling-day-dated with no confirmable time	script
E4	Result leakage — reports, cites, or reacts to the election’s outcome, exit information, or count, regardless of nominal date	human
E5	Irrelevant — fails every Local and National relevance rule (e.g. sport or entertainment merely naming a place)	human
E6	“Reform” false positive — matched only on “reform” in its ordinary sense, with no reference to the party	human
E7	Untrustworthy dating — only a crawl/capture or republication date is available and the original cannot be recovered	script
E8	Not editorial content — adverts, listings, notices, category/index, tag, or search-result pages	human
E9	Non-English	script
E10	Irretrievable — paywall, robots, or login wall blocks even an extract and no archive route succeeds	script

Table 2: Pre-registered article eligibility rules. Inclusion criteria I1–I6 and exclusion rules E1–E10 were applied by script or human review as specified. E4 prevented result leakage, and positive or negative coverage did not affect eligibility.

C LLM extraction validation

extraction layer	validation evidence	decision
Issue classification	$\kappa = 0.742$ between the two models; 0.616 against human labels (Sonnet, 53 articles)	passed; run with Sonnet 5
Stance (revised three-level)	$\kappa = 0.848$ between the two models (92 articles); 0.741 (Haiku) and 0.736 (Sonnet) against human labels (71 articles)	passed; run with Haiku 4.5
Framing: incumbent judgement	$\kappa = 0.705$ between the two models	passed, limited evidence
Framing: local impact	$\kappa = 0.635$ between the two models	passed, limited evidence
Framing: challenger emergence and voter discontent	only 2–6 and 4–5 positive articles among the 55 validation articles: too few to measure agreement	not assessable; exploratory use only
Expected impact	$\kappa = 0.598$ for both models after redesign (60 articles)	excluded, below 0.60
Credit–blame	$\kappa = 0.521 / 0.516$ against human labels (57 articles); a redesign did not improve it	excluded

Table 3: Final verification evidence and determination conclusions for each extraction layer; threshold $\kappa \geq 0.60$, with values representing the figures after validator gating.

D Residual-Based and Joint Design Comparison

news arm	window	residual design MAE	joint design MAE	baseline-only MAE	better design
combined	180-91 days	7.422	7.391	7.772	joint
combined	90-31 days	8.113	8.255	7.772	residual
combined	30-15 days	8.158	8.317	7.772	residual
combined	14-8 days	7.679	7.875	7.772	residual
combined	7-4 days	8.520	8.735	7.772	residual
combined	final 72h	9.102	9.239	7.772	residual
national	180-91 days	7.416	7.382	7.772	joint
national	90-31 days	8.115	8.257	7.772	residual
national	30-15 days	8.172	8.345	7.772	residual
national	14-8 days	7.653	7.819	7.772	residual
national	7-4 days	7.807	8.029	7.772	residual
national	final 72h	8.885	9.019	7.772	residual
local	180-91 days	7.598	7.820	7.772	residual
local	90-31 days	8.036	8.300	7.772	residual
local	30-15 days	7.767	8.000	7.772	residual
local	14-8 days	8.168	8.417	7.772	residual
local	7-4 days	8.520	8.735	7.772	residual
local	final 72h	7.813	8.019	7.772	residual

Table 4: Comparison of residual-based and joint models across 18 specifications, fitted on 2017 data and evaluated on 2021 data. This exploratory analysis was conducted after the final news-layer evaluation. The baseline-only column reports the Stage 1 baseline error without news features and serves as a reference.

E Stage 1 Predictors

predictor	meaning
analysis_number_of_seats	seats contested in the area
analysis_previous_turnout	turnout at the previous comparable election
area_parties_in_previous_contest	parties standing in the area's previous contest
authority	council holding the election
candidate_count_in_contest	candidates standing in the contest
candidate_previously_stood	candidate stood in an earlier election
contest_structure	single- or two-member contest
election_date	polling date
election_type	scheduled election or by-election
election_year	election year
first_appearance_of_party_in_area	party standing in this area for the first time
geographic_reference_eligibility	area has an approved exact-boundary historical reference
historical_predictor_availability	lagged predictors exist for this row
incumbent_candidate_yes_no	candidate holds the seat
incumbent_party_yes_no	party holds the seat
is_reform_uk	Reform UK indicator
is_ukip	UKIP indicator
party_candidate_count_in_contest	candidates the party fields in the contest
party_category	established, emerging, local or independent
party_contest_rate_previous	share of areas the party contested previously
party_contests_fought_previous	contests the party fought at the previous election
party_count_in_contest	parties standing in the contest
party_county_strength_previous	party's county-wide vote share at the previous election
party_county_strength_trend	change in county-wide vote share between elections
party_previously_contested	party stood in this area before
party_was_previous_winner	party won this area last time
previous_electorate	registered electors at the previous election
previous_party_vote_share	party's vote share in this area last time
previous_winning_party	party that won this area last time
reform_x_party_contest_rate_previous	Reform indicator x previous contest rate
reform_x_party_county_strength_previous	Reform indicator x previous county strength
reform_x_party_county_strength_trend	Reform indicator x county strength trend
reform_x_previous_party_vote_share	Reform indicator x previous area vote share
standard_party_name	standardised party label
years_since_previous_comparable_election	years since the previous comparable election

Table 5: The 35 original predictors used in the Stage 1 baseline model, taken from the published model feature schema. After encoding, these predictors formed 127 model input columns.

F Full Confirmatory Results

arm	window	control MAE	news MAE	Δ MAE	95% CI	Reform Δ
combined	91–180 days	4.441	4.753	-0.311	[-0.485, -0.144]	-0.446
combined	31–90 days	4.441	4.840	-0.399	[-0.590, -0.218]	-0.494
combined	15–30 days	4.441	5.357	-0.915	[-1.203, -0.638]	-1.272
combined	8–14 days	4.441	4.978	-0.536	[-0.779, -0.270]	-1.507
combined	4–7 days	4.441	6.635	-2.194	[-2.487, -1.896]	-2.184
combined	1–3 days	4.441	4.956	-0.515	[-1.137, +0.092]	-3.084
national	91–180 days	4.441	4.735	-0.294	[-0.477, -0.126]	-0.545
national	31–90 days	4.441	4.870	-0.429	[-0.642, -0.230]	-0.724
national	15–30 days	4.441	5.576	-1.135	[-1.625, -0.669]	-1.605
national	8–14 days	4.441	7.040	-2.598	[-2.909, -2.284]	-2.721
national	4–7 days	4.441	4.441	+0.000	[+0.000, +0.000]	+0.000
national	1–3 days	4.441	4.459	-0.018	[-0.031, -0.003]	-0.034

Table 6: Full v1 confirmatory results across combined and national news and the six predefined non-overlapping windows. Positive Δ MAE indicates lower prediction error with news than with the recalibrated control.

arm	window	control MAE	news MAE	Δ MAE	95% CI	Reform Δ
combined	91–180 days	4.445	5.021	-0.575	[-0.934, -0.245]	-2.805
combined	31–90 days	4.445	4.205	+0.240	[+0.078, +0.387]	-0.688
combined	15–30 days	4.445	4.312	+0.134	[+0.095, +0.169]	-0.148
combined	8–14 days	4.445	4.771	-0.325	[-0.569, -0.058]	-2.273
combined	4–7 days	4.445	4.455	-0.010	[-0.018, -0.002]	+0.020
combined	1–3 days	4.445	4.257	+0.188	[-0.230, +0.584]	-1.721
national	91–180 days	4.445	5.035	-0.590	[-0.947, -0.263]	-2.882
national	31–90 days	4.445	4.177	+0.268	[+0.109, +0.411]	-0.665
national	15–30 days	4.445	4.461	-0.016	[-0.124, +0.099]	-0.352
national	8–14 days	4.445	4.441	+0.005	[+0.001, +0.009]	-0.011
national	4–7 days	4.445	4.445	+0.000	[+0.000, +0.000]	+0.000
national	1–3 days	4.445	4.469	-0.023	[-0.039, -0.008]	+0.036

Table 7: Full v2 confirmatory results across combined and national news and the six predefined non-overlapping windows. Positive Δ MAE indicates lower prediction error with news than with the recalibrated control.

G Sensitivity Analysis Summary

version	fitted on	news arm	windows	n	improved	best Δ (window)	worst Δ
v1	2017 only	combined news	confirmed	6	0	+0.000 (1–3d)	-4.174
v1	2017 only	combined news	cumulative	6	1	+0.480 (prev 14d)	-4.660
v1	2017 only	local news	confirmed	6	0	+0.000 (1–3d)	-3.931
v1	2017 only	local news	cumulative	6	0	+0.000 (prev 72h)	-4.660
v1	2017 only	national news	confirmed	6	0	+0.000 (4–7d)	-6.270
v1	2017 only	national news	cumulative	6	0	+0.000 (prev 7d)	-6.270
v1	2017 + 2021	combined news	cumulative	6	2	+0.639 (prev 14d)	-0.515
v1	2017 + 2021	local news	confirmed	6	0	-0.107 (8–14d)	-2.194
v1	2017 + 2021	local news	cumulative	6	1	+0.320 (prev 7d)	-0.394
v1	2017 + 2021	national news	cumulative	6	0	-0.018 (prev 7d)	-3.245
v2	2017 + 2021 + by-elections	combined news	cumulative	6	5	+0.610 (prev 14d)	-0.391
v2	2017 + 2021 + by-elections	local news	confirmed	6	3	+0.289 (91–180d)	-0.833
v2	2017 + 2021 + by-elections	local news	cumulative	6	5	+0.508 (prev 14d)	-0.833
v2	2017 + 2021 + by-elections	national news	cumulative	6	1	+0.081 (prev 90d)	-1.739

Table 8: Summary of the sensitivity analyses across news types and settings, reporting the number of comparisons, the number of MAE improvements, and the range of Δ MAE relative to the adjustment control. In the windows column, “confirmed” refers to the six predefined non-overlapping pre-election windows in Section 3.2.1, while “cumulative” refers to six overlapping windows extending backward from polling day.

H Local News Sensitivity Analysis

window	recal. MAE	news MAE	Δ MAE	recal. RMSE	news RMSE
91–180 days	4.4454	4.1560	+0.2894	5.7538	5.4518
31–90 days	4.4454	4.2571	+0.1884	5.7538	5.3110
15–30 days	4.4454	5.0069	−0.5614	5.7538	6.0874
8–14 days	4.4454	4.4271	+0.0184	5.7538	5.7494
4–7 days	4.4454	4.4551	−0.0097	5.7538	5.7500
final 72h	4.4454	5.2784	−0.8329	5.7538	6.8172

Table 9: Sensitivity analysis using local news features. The table reports MAE and RMSE for local news across the six non-overlapping windows, with Δ MAE measured relative to the v2 no-news control (MAE 4.4454); positive values indicate lower error with news.

I Cumulative-Window Results

news arm	window	news MAE	Δ MAE	news RMSE	Reform Δ
combined	previous 72 hours	4.2572	+0.1882	5.3666	-1.7214
combined	previous 7 days	4.0324	+0.4130	5.1499	-1.3281
combined	previous 14 days	3.8353	+0.6101	4.8741	+0.0659
combined	previous 30 days	4.1823	+0.2632	5.2824	-0.2499
combined	previous 90 days	4.2513	+0.1942	5.2685	-0.7549
combined	previous 180 days	4.8365	-0.3910	5.7988	-2.5994
national	previous 72 hours	4.4686	-0.0231	5.7476	+0.0360
national	previous 7 days	4.4686	-0.0231	5.7476	+0.0360
national	previous 14 days	4.4563	-0.0109	5.7496	+0.0219
national	previous 30 days	6.1846	-1.7391	7.6656	-2.9014
national	previous 90 days	4.3643	+0.0812	5.3527	-1.1610
national	previous 180 days	4.9258	-0.4804	5.8890	-2.8510
local	previous 72 hours	5.2784	-0.8329	6.8172	-3.8680
local	previous 7 days	4.0404	+0.4050	5.1202	-1.1014
local	previous 14 days	3.9373	+0.5081	5.1004	-0.1378
local	previous 30 days	4.1623	+0.2832	5.2813	-0.1483
local	previous 90 days	4.3635	+0.0819	5.3603	-0.2112
local	previous 180 days	4.3493	+0.0962	5.2989	-0.1123

Table 10: Cumulative-window results for all 18 comparisons(three news arms: local, national and combined across six non-overlapping windows). Δ MAE is measured relative to the v2 no-news control MAE of 4.4454, with positive values indicating lower error with news. The Reform UK column reports the corresponding Δ MAE for Reform UK candidates only.

J Leave-One-Party-Out Sensitivity Analysis

party omitted	comparisons	improvements	coefficient sign flips
Conservative	18	6	3
Green	18	0	4
Labour	18	0	2
Liberal Democrat	18	0	5
UKIP	18	0	4
Total	90	6	—

Table 11: Leave-one-party-out sensitivity analysis. The table reports the number of MAE improvements across the 18 comparisons(three news arms: local, national and combined across six non-overlapping windows) when each party is removed from the training sample.

K Reform UK Training Cell Attribution Analysis

arm	window	full-fit Δ	no-Reform Δ	attribution
combined	91–180 days	−0.5751	−1.2684	no full-fit improvement
combined	31–90 days	+0.2404	+0.0773	improvement retained without Reform cells
combined	15–30 days	+0.1339	+0.0333	improvement retained without Reform cells
combined	8–14 days	−0.3252	−0.8817	no full-fit improvement
combined	4–7 days	−0.0097	−0.0010	no full-fit improvement
combined	final 72h	+0.1882	−1.1328	improvement lost without Reform cells
national	91–180 days	−0.5896	−1.2782	no full-fit improvement
national	31–90 days	+0.2682	−0.0015	improvement lost without Reform cells
national	15–30 days	−0.0156	−2.6482	no full-fit improvement
national	8–14 days	+0.0048	+0.0029	improvement retained without Reform cells
national	4–7 days	0.0000	0.0000	no full-fit improvement
national	final 72h	−0.0231	−0.0050	no full-fit improvement

Table 12: Post-evaluation exploratory attribution of Δ MAE to Reform UK training cells. The table compares the full-fit model with 45 training cells, including seven Reform UK cells, against a no-Reform specification with 38 cells across combined and national news settings and prediction windows. The difference between specifications represents the contribution of the Reform UK training cells. Positive Δ MAE values indicate that news reduced prediction error.

L Seat-call Accuracy Results

arm	window	seat-call accuracy (news)	vs control	beats 0.8053 floor
combined	91–180 days	0.8317	+0.1129	yes
combined	31–90 days	0.7909	+0.0721	no
combined	15–30 days	0.7212	+0.0024	no
combined	8–14 days	0.6827	-0.0361	no
combined	4–7 days	0.7188	+0.0000	no
combined	1–3 days	0.8558	+0.1370	yes
national	91–180 days	0.8317	+0.1129	yes
national	31–90 days	0.7909	+0.0721	no
national	15–30 days	0.7212	+0.0024	no
national	8–14 days	0.7188	+0.0000	no
national	4–7 days	0.7188	+0.0000	no
national	1–3 days	0.7188	+0.0000	no

Table 13: Seat-call accuracy for each v2 specification and its difference from the non-news adjustment control. The final column shows whether each specification exceeded the 0.8053 elect-nobody benchmark; seat-call accuracy remained a secondary outcome and was not promoted to a confirmatory finding.

M Transfer and Replication Test Results

window	arm	Woking South		Haslemere	
		news MAE	Δ	news MAE	Δ
91–180 days	combined	9.366	+0.719	8.418	-3.926
91–180 days	local*	13.889	-3.804	–	–
91–180 days	national	9.212	+0.873	4.557	-0.066
31–90 days	combined	10.309	-0.225	4.290	+0.201
31–90 days	local	9.917	+0.168	–	–
31–90 days	national*	10.337	-0.252	4.580	-0.089
15–30 days	combined*	10.374	-0.289	4.596	-0.105
15–30 days	local	9.770	+0.315	–	–
15–30 days	national	10.390	-0.305	4.603	-0.112
8–14 days	combined	9.998	+0.087	4.437	+0.054
8–14 days	local*	10.011	+0.073	–	–
8–14 days	national	10.082	+0.003	4.472	+0.019
4–7 days	combined	10.212	-0.127	4.527	-0.036
4–7 days	local	10.212	-0.127	–	–
4–7 days	national*	10.126	-0.041	4.491	+0.000
1–3 days	combined	10.299	-0.214	4.564	-0.073
1–3 days	local*	10.012	+0.072	–	–
1–3 days	national	10.301	-0.216	4.565	-0.074

Table 14: Full results for the Woking South and Haslemere transfer tests. Woking South results are measured relative to the no-news baseline (MAE 10.085), and Haslemere results are measured relative to the non-news adjustment control (MAE 4.491). The asterisk marks the specification selected by the pre-registered rule as the arm with the highest 2026 Δ MAE in each window. Haslemere replayed only the two confirmatory arms, combined and national news, so local entries are shown as “–”. Its news corpus contained 20 articles, but only three mentioned a study party.

N Exploratory Issue and Framing Features

arm	Δ MAE, 91–180 days	Δ MAE, 31–90 days	CI above bar
volume + stance (the bar)	+0.1892 [-0.0504, +0.4110]	+0.3413 [+0.2023, +0.4680]	—
framing: incumbent judgement	-0.6094 [-0.9676, -0.2810]	+0.5718 [+0.3864, +0.7533]	yes
issue: other	-0.5767 [-0.9474, -0.2349]	+0.3446 [+0.1856, +0.4857]	no
issue: immigration	+0.0958 [-0.1418, +0.3274]	+0.3202 [+0.1083, +0.5119]	no
framing: challenger emergence	-0.5047 [-0.8559, -0.1826]	+0.3167 [+0.1192, +0.5022]	no
framing: local impact	-0.6554 [-0.9836, -0.3428]	+0.2903 [+0.1326, +0.4374]	no
issue: national politics	-0.5763 [-0.9354, -0.2455]	+0.2050 [+0.0687, +0.3275]	no
framing: voter discontent	-0.5758 [-0.9345, -0.2455]	+0.1761 [+0.0095, +0.3331]	no
all seven at once	-0.1071 [-0.3490, +0.1263]	-0.6635 [-0.8436, -0.4841]	no

Table 15: Exploratory results for seven issue and framing features defined before fitting. Each was added separately to the frozen specification. In the 31–90 day window, the +0.3413 volume-plus-stance improvement was used as the reference. The final row tests all seven features together for overfitting. Only the 91–180 and 31–90 day windows met the ten-value reporting threshold. At 91–180 days, six feature effects were negative, matching the frozen news effect, with confidence intervals below zero. The immigration effect was the exception, with a confidence interval including zero.

O Stage 2 Training Data and Test Timeline

election	polling date	news cut-off	party	Stage 1 rows	role
By-election Weybridge	2015-05-07	2015-05-06	Conservative	1	training
By-election Weybridge	2015-05-07	2015-05-06	Labour	1	training
By-election Weybridge	2015-05-07	2015-05-06	UKIP	1	training
SCC 2017	2017-05-04	2017-05-03	Conservative	81	training
SCC 2017	2017-05-04	2017-05-03	Green	55	training
SCC 2017	2017-05-04	2017-05-03	Labour	75	training
SCC 2017	2017-05-04	2017-05-03	Liberal Democrats	78	training
SCC 2017	2017-05-04	2017-05-03	UKIP	58	training
By-election Warlingham	2019-01-31	2019-01-30	Conservative	1	training
By-election Warlingham	2019-01-31	2019-01-30	Labour	1	training
By-election Warlingham	2019-01-31	2019-01-30	Liberal Democrats	1	training
By-election Warlingham	2019-01-31	2019-01-30	UKIP	1	training
SCC 2021	2021-05-06	2021-05-05	Conservative	81	training
SCC 2021	2021-05-06	2021-05-05	Green	36	training
SCC 2021	2021-05-06	2021-05-05	Labour	79	training
SCC 2021	2021-05-06	2021-05-05	Liberal Democrats	72	training
SCC 2021	2021-05-06	2021-05-05	Reform UK	6	training
SCC 2021	2021-05-06	2021-05-05	UKIP	5	training
By-election Nork Tattenhams	2025-05-01	2025-04-30	Conservative	1	training
By-election Nork Tattenhams	2025-05-01	2025-04-30	Green	1	training
By-election Nork Tattenhams	2025-05-01	2025-04-30	Labour	1	training
By-election Nork Tattenhams	2025-05-01	2025-04-30	Liberal Democrats	1	training
By-election Nork Tattenhams	2025-05-01	2025-04-30	Reform UK	1	training
By-election Addlestone	2025-08-21	2025-08-20	Conservative	1	training
By-election Addlestone	2025-08-21	2025-08-20	Green	1	training
By-election Addlestone	2025-08-21	2025-08-20	Labour	1	training
By-election Addlestone	2025-08-21	2025-08-20	Liberal Democrats	1	training
By-election Addlestone	2025-08-21	2025-08-20	Reform UK	1	training
By-election Hinchley Wood C. Oxshott	2025-08-21	2025-08-20	Green	1	training
By-election Hinchley Wood C. Oxshott	2025-08-21	2025-08-20	Labour	1	training
By-election Hinchley Wood C. Oxshott	2025-08-21	2025-08-20	Liberal Democrats	1	training
By-election Hinchley Wood C. Oxshott	2025-08-21	2025-08-20	Reform UK	1	training
By-election Camberley West	2025-10-16	2025-10-15	Labour	1	training
By-election Camberley West	2025-10-16	2025-10-15	Liberal Democrats	1	training
By-election Camberley West	2025-10-16	2025-10-15	Reform UK	1	training
By-election Caterham Valley	2025-10-16	2025-10-15	Conservative	1	training
By-election Caterham Valley	2025-10-16	2025-10-15	Green	1	training
By-election Caterham Valley	2025-10-16	2025-10-15	Labour	1	training
By-election Caterham Valley	2025-10-16	2025-10-15	Liberal Democrats	1	training
By-election Caterham Valley	2025-10-16	2025-10-15	Reform UK	1	training
By-election Guildford South East	2025-10-16	2025-10-15	Conservative	1	training
By-election Guildford South East	2025-10-16	2025-10-15	Green	1	training
By-election Guildford South East	2025-10-16	2025-10-15	Labour	1	training
By-election Guildford South East	2025-10-16	2025-10-15	Liberal Democrats	1	training
By-election Guildford South East	2025-10-16	2025-10-15	Reform UK	1	training
2026 East and West Surrey (pooled)	2026-05-07	2026-05-06	all six parties	832	frozen test

Table 16: Stage 2 fitting cells and the frozen 2026 test set. All training elections pre-date the June 2026 project start. The v1 and v2 freeze commits precede the final news-layer unblinding commit, as verified by the committed repository audit.